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ALLIANCE UNIVERSITY

Alliance School of Advanced Computing

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Project Title:

Analysis of Crimes Against Scheduled Castes in India (2021–2023)

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1. Abstract

This project analyses state- and union-territory-level records of crimes against Scheduled Castes (SCs) for 2021, 2022 and 2023 using the supplied NCRB CII 2023 Table 7A.1 CSV. The dataset includes yearly case counts, the 2011 SC population field in lakhs, a 2023 crime-rate field and a 2023 chargesheeting-rate field. The analysis focuses on data inspection, cleaning, comparison across jurisdictions, year-to-year change, and communicating patterns through charts and a Power BI report artifact.

The uploaded CSV contains 39 rows and 8 columns, including aggregate rows. After excluding aggregate total rows, 36 State/UT records remain for jurisdiction-level comparisons. The sum across these jurisdiction rows is 50,900 for 2021, 57,582 for 2022 and 57,789 for 2023. These sums are computed from the supplied rows and should not be treated as a replacement for an official national total without checking the source table's aggregation conventions.

The project is descriptive, not causal. Recorded case counts do not independently measure prevalence, risk, reporting behaviour, conviction outcomes or the quality of justice. Interpretation should consider population, reporting, legal definitions and source-table notes. The supplied notebook also refers to a different dataset schema (2020–2022 cognizable crimes), so it cannot be assumed to have executed against the uploaded 2021–2023 SC-specific CSV without adaptation.

2. Introduction

Data analysis can help organize official crime statistics into comparable tables, charts and summary measures. This project uses a state/UT table concerning crimes against Scheduled Castes in India. The intended audience is an academic reader who needs to understand how a public statistical dataset can be inspected, cleaned, analysed and presented in a dashboard.

The CSV contains historical case counts for three years and two additional rate fields for 2023. It also includes a population field for Scheduled Castes, measured in lakhs and sourced from the 2011 population reference in the column heading. The difference in reference years matters: the population field is not a 2023 population estimate.

The report preserves the supplied dataset's terminology and distinguishes reported cases from rates. It does not infer why a count changes or assign a judgement to any State/UT. Where a field is missing, the report identifies that limitation rather than silently imputing a value.

2.1 Project Title

Analysis of Crimes Against Scheduled Castes in India (2021–2023) Using NCRB Data.

2.2 Purpose

• Summarize the supplied dataset and its fields.

• Compare reported case counts across the three years.

• Identify jurisdictions with higher counts in the supplied 2023 column.

• Document data-quality and notebook-schema issues.

• Present findings in a form suitable for academic review and Power BI exploration.

3. Problem Statement and Objectives

3.1 Problem Statement

Crime statistics may be difficult to compare when values are spread across tables or viewed without consistent definitions. A structured analytical workflow is needed to check data quality, compare years and jurisdictions, and communicate findings while avoiding unsupported conclusions.

3.2 Objectives

1. Load and inspect the provided NCRB CII Table 7A.1 CSV.

2. Check the dimensions, data types, missing values, duplicate records and aggregate rows.

3. Compare the 2021, 2022 and 2023 reported case-count columns.

4. Calculate absolute change between 2021 and 2023 for descriptive comparison.

5. Visualize the highest reported 2023 counts among State/UT rows.

6. Review the notebook and Power BI artifact for alignment with the CSV.

7. Document limitations and recommendations for a reproducible analysis.

3.3 Scope

The scope is limited to the uploaded CSV, Jupyter notebook and Power BI report archive. No individual case records are supplied. This report therefore analyses aggregate jurisdiction-level values only and does not attempt person-level, district-level or incident-level analysis.

4. Existing System

A conventional review of an official statistical table often relies on manually reading rows, comparing yearly columns and preparing charts separately. This can be adequate for a quick review but may be difficult to reproduce when multiple comparisons or updates are needed.

4.1 Common Challenges

• Aggregate totals can be accidentally mixed with State/UT records.

• A chart may be misleading if the axis, year or measure is not labelled clearly.

• Missing rate fields can be mistaken for zero values.

• Different source-table schemas may be confused when notebooks are reused.

• Counts and rates answer different questions and should not be interpreted interchangeably.

• A dashboard may show a visually clear result without exposing the data-cleaning assumptions.

4.2 Need for Structured Analysis

A documented workflow makes it easier to trace every chart back to the CSV, identify how aggregate rows were handled, and repeat calculations when the source table is updated. It also helps an evaluator distinguish what is directly supported by the source from what remains an interpretation.

5. Proposed System

The proposed analytical workflow consists of source-data ingestion, validation, cleaning, derived calculations, exploratory analysis, visualisation, and reporting. The CSV remains the source for the numeric results in this report. The notebook is reviewed as a methodological artefact, while the Power BI archive is documented as a dashboard/report artefact.

5.1 Main Modules

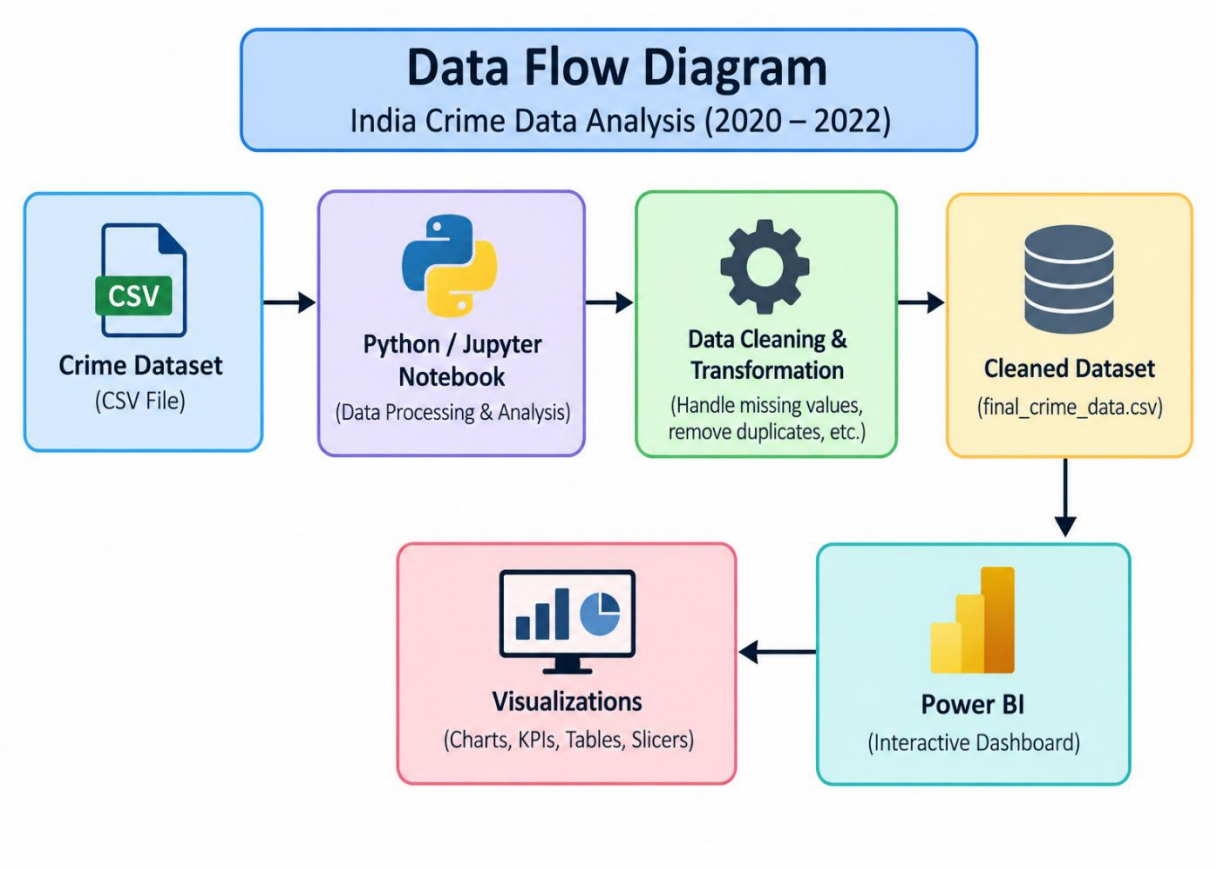
Module	Purpose
Data ingestion	Load the supplied CSV and preserve original columns.
Data validation	Inspect row counts, field types, missing values and duplicate rows.
Aggregate-row handling	Separate State/UT records from national or grouped total rows.
Descriptive analysis	Summarize counts and year-to-year differences.
Visualization	Create labelled charts for comparison and trend review.
Notebook review	Check that notebook columns match the actual dataset.
Power BI report	Provide a visual interface for exploring the data model and report pages.
Documentation	Record methods, assumptions, limitations and source files.

5.2 Workflow

1. Load NCRB_CII_2023_Table_7A.1_0.csv.
2. Inspect column names and data types.
3. Convert numeric fields to numeric types while preserving missing values.
4. Identify and exclude aggregate rows for jurisdiction-only comparisons.
5. Calculate 2021–2023 absolute and percentage changes where valid.
6. Create charts and summary tables.
7. Review the notebook's expected schema against the CSV.

8. Inspect the Power BI report archive structure.

9. Write source-grounded conclusions and document unresolved gaps.



6. Dataset Description

The uploaded file has 39 rows and 8 columns. The State/UT field includes 36 jurisdiction-level rows after removing aggregate rows used for national or grouped totals. The original CSV includes the following fields:

Field	Description
Sl. No.	Serial number in the source table.
State/UT	State or Union Territory label.
2021	Reported case count for 2021.
2022	Reported case count for 2022.
2023	Reported case count for 2023.
Actual Population of SCs (in Lakhs) (2011)	SC population field referenced to 2011, in lakhs.
Rate of Total Crime against SCs (2023)	Rate field for 2023 as named in the source CSV.

Chargesheeting Rate (2023)	Chargesheeting-rate field for 2023 as named in the source CSV.

6.1 Dataset Dimensions and Notes

Check	Result
Raw rows	39
Raw columns	8
State/UT records used for comparisons	36
Missing crime-count values	0
Missing 2023 crime-rate values	4
Missing 2023 chargesheeting-rate values	9

Missing rate values are retained as missing and are not converted to zero. The 2011 population field and 2023 rates should be interpreted using the official source table's definitions and footnotes.

7. Data Cleaning and Preparation

The report reads the CSV as supplied and converts year and rate columns to numeric values for calculation. Aggregate rows are excluded from State/UT comparisons so that summary rows do not distort the ranking of individual jurisdictions.

7.1 Preparation Steps

1. Read the CSV while preserving its original column labels.
2. Inspect missing values and duplicates.
3. Normalize State/UT labels for identifying aggregate rows.
4. Exclude aggregate rows from jurisdiction-level charts and rankings.
5. Calculate absolute change as 2023 minus 2021.
6. Calculate percentage change only where the 2021 value is greater than zero.
7. Keep missing rate values missing and disclose them in the report.
8. Retain the original CSV unchanged as the source input.

7.2 Data Quality Considerations

• A zero case count and a missing rate are not the same thing.

• Percentage change is undefined when the baseline count is zero.

• The population field is labelled with the 2011 reference year.

• The source table may contain specific footnotes or aggregation rules that should be consulted before publication.

• Data cleaning should be reproducible and should not silently overwrite original values.

8. Exploratory Data Analysis

Across the 36 jurisdiction-level rows used in this report, the sum of the case-count column is 50,900 in 2021, 57,582 in 2022 and 57,789 in 2023. The 2023 sum is 6,889 higher than the 2021 sum across these rows, equivalent to 13.53% relative to the 2021 sum.

These figures are computed by summing the State/UT rows in the uploaded CSV after excluding aggregate rows. They are descriptive calculations and should be checked against the official publication's national totals and table notes before being quoted as official totals.

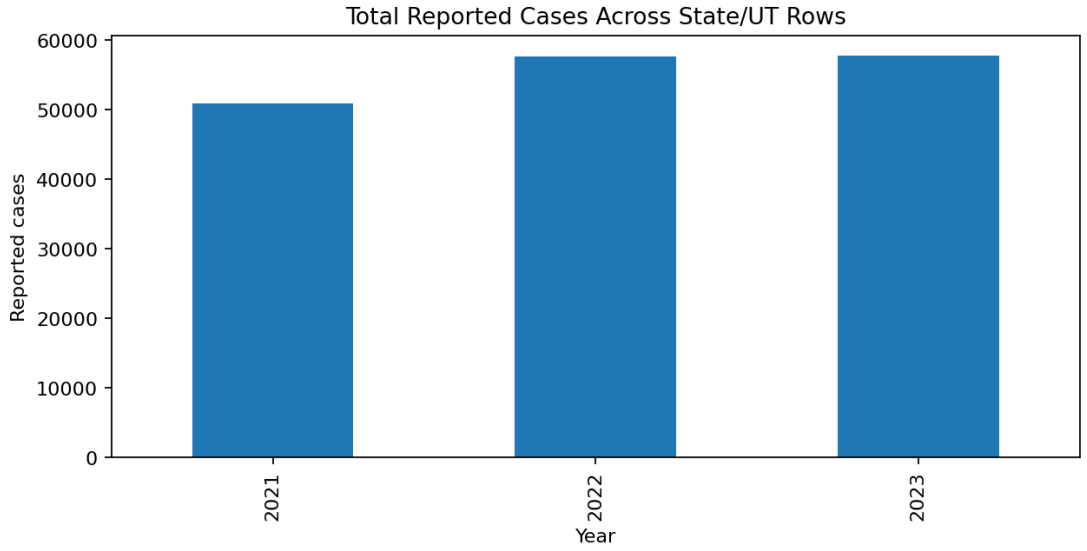


Figure 1. Sum of reported case counts across jurisdiction-level rows, 2021–2023.

8.1 Interpretation

The increase between the 2021 and 2022 sums is larger than the change between 2022 and 2023 in the supplied rows. The aggregate pattern does not imply that every jurisdiction followed the same direction; individual changes vary.

9. State/UT Comparison

The 2023 column can be used to rank the State/UT rows by reported case count. The top entries in the supplied CSV are Uttar Pradesh, Rajasthan, Madhya Pradesh and Bihar. Rankings by raw counts should not be interpreted as rankings by population-adjusted rate or individual risk.

Rank	State/UT	2021	2022	2023
1	Uttar Pradesh	13146	15368	15130
2	Rajasthan	7524	8752	8449
3	Madhya Pradesh	7214	7733	8232
4	Bihar	5842	6509	7064
5	Maharashtra	2503	2743	3024
6	Odisha	2327	2902	2696
7	Andhra Pradesh	2014	2315	2027
8	Karnataka	1673	1977	1923
9	Tamil Nadu	1377	1761	1921
10	Telangana	1772	1787	1709

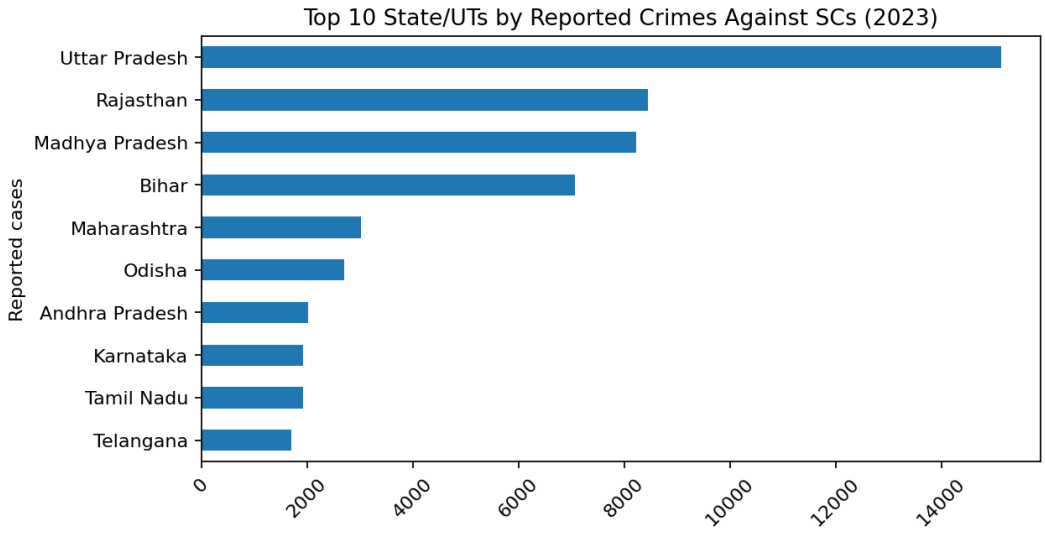


Figure 2. Ten State/UT rows with the highest reported 2023 case counts.

A count is affected by population size, reporting and other contextual factors. If the analytical question concerns relative rate rather than total case volume, the rate field and its denominator definition must be used instead, with missing values treated carefully.

10. Trend Analysis (2021–2023)

The year-to-year columns enable two kinds of descriptive comparison: absolute change (difference in number of cases) and percentage change relative to a baseline. Percentage change is only calculated when the baseline year has a positive count. A percentage should not be reported for a zero baseline because the division is undefined.

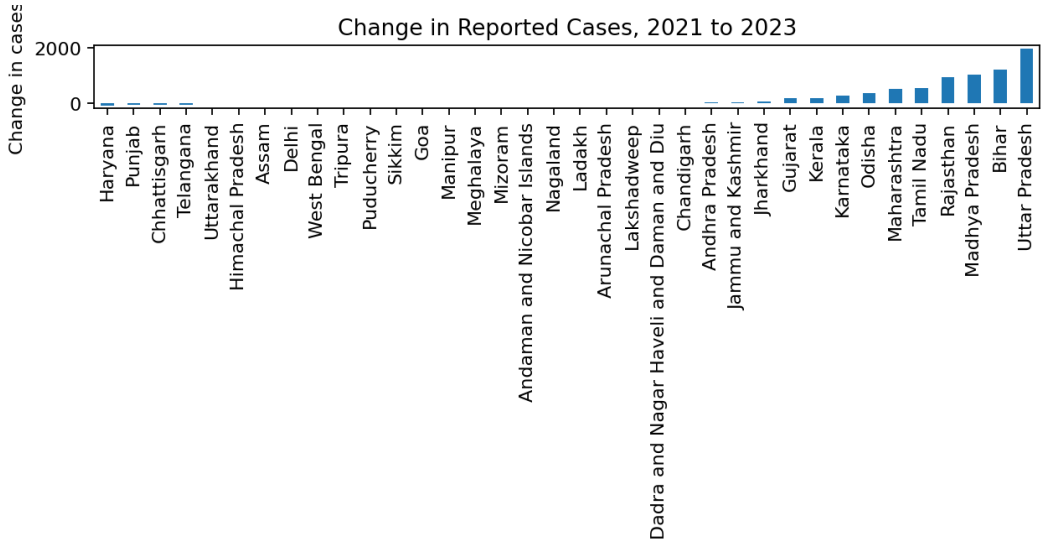


Figure 3. Absolute change in reported case counts between 2021 and 2023 across State/UT rows.

10.1 Change Summary

Pattern from 2021 to 2023	Number of State/UT rows
Increase	15
Decrease	12
No change	9

The direction of change is not uniform across jurisdictions. These counts describe the uploaded table only and do not explain the underlying reasons for change.

11. Crime Rate and Chargesheeting Fields

The dataset includes a 2023 rate of total crime against Scheduled Castes and a 2023 chargesheeting-rate field. These measures are distinct from raw case counts and must be interpreted according to the definitions and denominators in the official NCRB table.

11.1 Missing Values

The CSV contains 4 missing values in the 2023 crime-rate field and 9 missing values in the 2023 chargesheeting-rate field. These are not treated as zero in this report.

11.2 Interpretation Caution

• Do not compare rates unless the numerator, denominator and reference population are consistent.

• Do not interpret a chargesheeting rate as a conviction rate.

• Do not infer case outcomes from the case-count columns.

• Consult the original NCRB table notes for precise definitions and exceptions.

12. Notebook Review and Methodology

The uploaded Jupyter notebook contains code for loading a file named NCRB_Table_1A.1.csv, excluding aggregate rows, analysing columns named 2020, 2021 and 2022, and deriving a crime-change variable for 2020–2022. It also refers to fields named 'Mid-Year Projected Population (in Lakhs) (2022)', 'Rate of Cognizable Crimes (IPC) (2022)' and 'Chargesheeting Rate (2022)'.

These expected fields do not match the supplied CSV, which is titled NCRB_CII_2023_Table_7A.1_0.csv and contains 2021, 2022 and 2023 columns plus SC-specific 2023 rate fields. Therefore, the notebook cannot be described as having successfully run against the supplied CSV without code changes. The report's numeric findings are calculated directly from the uploaded CSV instead.

12.1 Notebook Methods Evidenced in Source

• Pandas for loading and manipulating tabular data.

• NumPy for numeric transformations.

• Matplotlib and Seaborn imports for visualisation.

• Scikit-learn is installed/imported in setup cells, but the visible notebook code focuses primarily on descriptive analysis and outlier detection.

• IQR-based outlier detection is applied to a crime-count column in the notebook's expected schema.

• The notebook writes cleaned and final CSV outputs in its execution environment.

12.2 Required Adaptation

To make the notebook reproducible with this CSV, update the input path and select the 2021, 2022 and 2023 fields; revise all renamed population/rate fields to the SC-specific columns; update the change calculations to 2021–2023; and rerun every cell from a clean kernel.

13. Power BI Report Overview

The supplied archive 'Crime Analysis.pbix.zip' contains a Power BI report-definition structure, theme files, page definitions, visual definitions and a data-model artefact. The archive appears to contain a report project structure rather than a conventional single .pbix file at its root, so the archive format and opening workflow should be checked in Power BI Desktop before presenting it as a directly openable PBIX.

The report structure includes multiple report pages and visual definitions. Because the supplied archive does not expose a simple exported screenshot or readable final chart images, this report documents the artifact structure but does not claim specific Power BI chart values beyond those calculated from the CSV.

13.1 Suggested Dashboard Areas

• Year-wise reported case counts.

• State/UT comparison for 2023.

• Absolute change from 2021 to 2023.

• 2023 crime-rate field with missing values visibly indicated.

• 2023 chargesheeting-rate field with its own definition and missingness.

• Tooltips for source field definitions and data caveats.

14. System Architecture and Data Flow

Layer	Input / Component	Role
Source	NCRB CII CSV	Original State/UT case counts and rate fields.
Preparation	Python, Pandas, NumPy	Load, validate, clean and derive comparison columns.
Analysis	Descriptive calculations	Summarize counts, changes and missingness.
Visualization	Matplotlib / Power BI	Create static figures and interactive views.
Review	Jupyter notebook	Document transformations and rerun calculations.
Output	Report, CSV exports and dashboard	Communicate findings and allow further exploration.

14.1 Data Flow

1. Read the original CSV.

2. Validate the columns and missing values.

3. Separate aggregate rows from jurisdiction rows.

4. Calculate trend and ranking measures.

5. Generate charts and tables.

6. Review notebook/report consistency.

7. Present the analysis with caveats and source references.

The data flow is analytical rather than a live operational pipeline. No live database, API, automatic refresh service or production deployment is established by the uploaded files.

15. Output and Findings

15.1 Key Findings

• The source CSV has 39 rows and 8 columns before filtering.

• 36 State/UT records were used for jurisdiction comparisons after removing aggregate rows.

• The summed jurisdiction-level count is 50,900 for 2021, 57,582 for 2022 and 57,789 for 2023.

• Uttar Pradesh has the highest 2023 count in the supplied State/UT rows (15,130), followed by Rajasthan (8,449).

• From 2021 to 2023, 15 rows increased, 12 decreased and 9 remained unchanged.

•The 2023 rate fields contain missing values, so rate comparisons should disclose data availability.

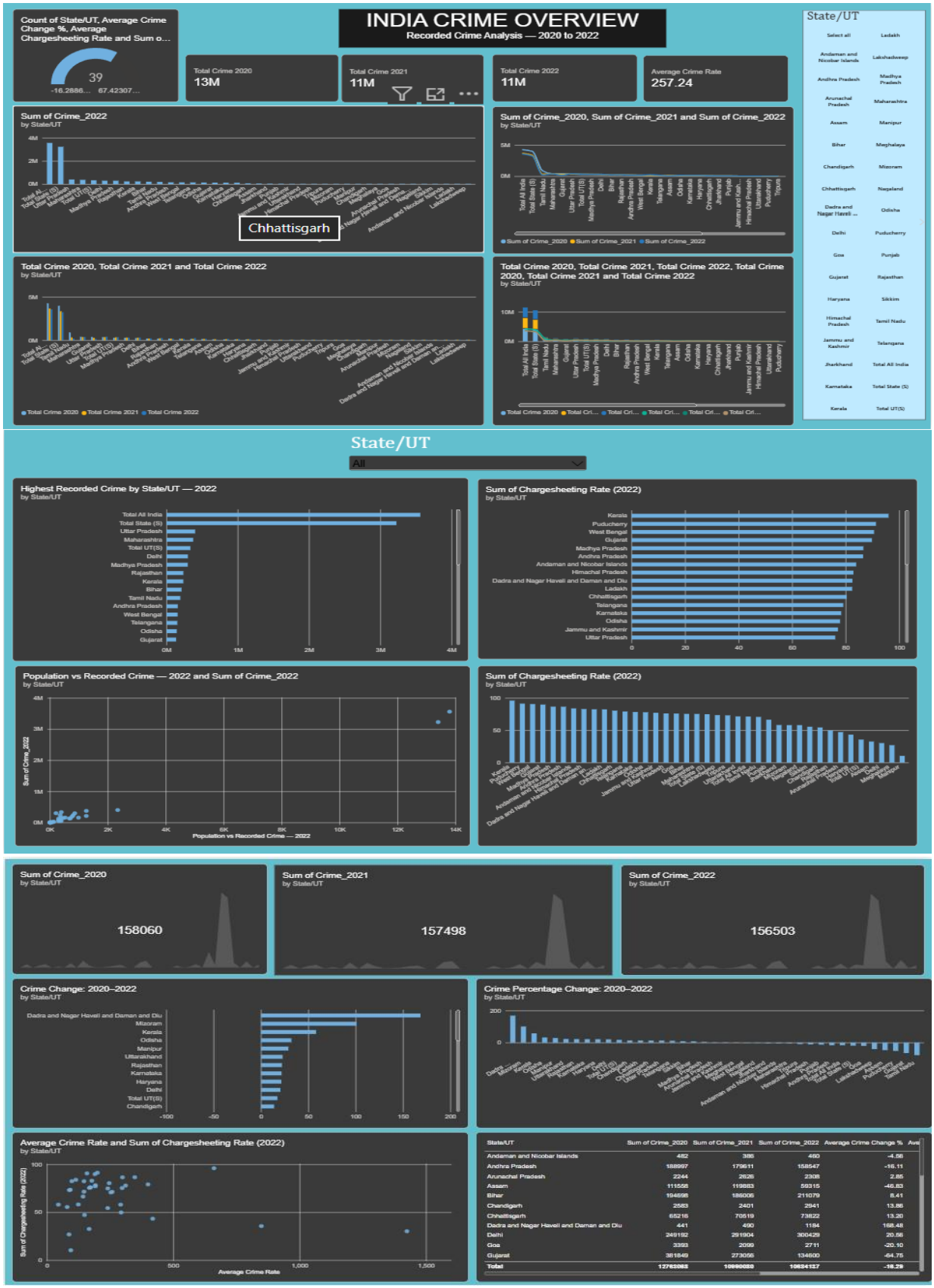


Figure 4. Highest 12 State/UT rows by reported case count in 2023.

16. Testing and Validation

The following are recommended validation checks for a repeatable academic workflow; they are not a claim that every test was executed in the supplied notebook.

Test	Validation	Expected Outcome
CSV loading	Read the exact supplied path.	File loads without column substitution.
Schema check	Compare expected and actual columns.	Notebook reports mismatch clearly.
Aggregate rows	Check labels for total rows.	Aggregate rows excluded from jurisdiction rankings.
Numeric types	Check year and rate columns.	Calculations use numeric values.
Missingness	Count null values in rate fields.	Missing values remain missing.
Change formula	Manually check selected rows.	2023 minus 2021 is reproduced correctly.
Zero baseline	Calculate percentage change where 2021=0.	Percentage is marked undefined.
Chart labels	Review title, axis and year.	Measure and time period are clear.
Reproducibility	Restart kernel and run all cells.	No stale variables or machine-specific path errors.

17. Limitations and Responsible Interpretation

- The CSV contains aggregate statistics, not individual case records.
- The population field references 2011, while the case counts cover 2021–2023.
- The notebook's expected input schema differs from the uploaded CSV.
- The report does not establish causation or explain changes in counts.
- Raw counts are not population-adjusted rates.
- Missing rate fields limit comparisons.
- The Power BI archive's precise opening/refresh workflow is not confirmed.
- No predictive model or validated forecast is evidenced by the supplied files.

Crime statistics can be affected by recording practices, reporting behaviour, legal classifications and institutional processes. The analysis should be used for descriptive academic exploration, not to stereotype communities or infer the conduct of individuals.

18. Future Scope

• Align the notebook's input schema with the uploaded SC-specific dataset.

• Add automated schema validation and clear error messages.

• Create reproducible preprocessing and output folders.

• Document the official source table's definitions and footnotes.

• Build Power BI measures for year-wise counts and valid rate comparisons.

• Add filters for year and State/UT while keeping aggregate totals separate.

• Add a data dictionary and source/version metadata.

• Validate all dashboard calculations against independent CSV summaries.

• If adding forecasting, use a suitable time series with sufficient history and validate it out of sample.

19. Conclusion

This project demonstrates a descriptive analysis workflow for the supplied NCRB CII Table 7A.1 dataset concerning crimes against Scheduled Castes. The CSV supports comparison of reported case counts across 2021, 2022 and 2023 and includes 2023 rate fields. The report provides data-quality summaries, state/UT rankings, year-to-year change calculations and visualisations.

The analysis also identifies an important reproducibility issue: the supplied notebook expects a different NCRB dataset and older year columns. The report therefore calculates numeric findings from the uploaded CSV rather than presenting the notebook as already compatible. The Power BI archive provides a report-definition/model artefact, but its exact opening and refresh workflow should be verified.

The project is suitable as an academic descriptive-analysis prototype. Further work should prioritise source-definition verification, notebook schema alignment, repeatable data validation and dashboard consistency. Findings should remain descriptive and should not be treated as explanations of crime or justice outcomes.

20. References and Source Materials

1. NCRB_CII_2023_Table_7A.1_0.csv — uploaded State/UT-level dataset containing case counts for 2021–2023 and two 2023 rate fields.

2. a319156e-8e70-4445-97c6-5561b062edfc.ipynb — uploaded notebook containing data loading, cleaning, descriptive analysis, renaming, outlier checks and CSV export code.

3. Crime Analysis.pbix.zip — uploaded Power BI report-definition archive containing theme, page, visual and data-model files.

4. Official NCRB source-table documentation and footnotes should be consulted to verify field definitions and aggregate totals before external publication.

Appendix A. Selected 2023 Data

State/UT	2023 reported cases
Uttar Pradesh	15130
Rajasthan	8449
Madhya Pradesh	8232
Bihar	7064
Maharashtra	3024
Odisha	2696
Andhra Pradesh	2027
Karnataka	1923
Tamil Nadu	1921
Telangana	1709
Haryana	1539
Gujarat	1373
Kerala	1128
Jharkhand	604
Chhattisgarh	250

Appendix B. Reproducibility Checklist

- Keep an unchanged copy of the original CSV.
- Record file name, date and source table.
- Run schema checks before analysis.
- Keep aggregate rows separate from State/UT rows.
- Recalculate figures after each source update.
- Keep missing values distinct from zero.
- Verify Power BI measures against the CSV.
- Record notebook changes and rerun from a clean kernel.

Appendix C. Data Dictionary and Derived Measures

Measure	Formula / Source	Interpretation
Reported cases 2021	CSV field '2021'	Source-reported count for the year.
Reported cases 2022	CSV field '2022'	Source-reported count for the year.
Reported cases 2023	CSV field '2023'	Source-reported count for the year.
Absolute change	2023 count − 2021 count	Positive indicates an increase in recorded count.
Percentage change	(2023 − 2021) / 2021 × 100	Calculated only when 2021 is greater than zero.
2023 crime rate	CSV field 'Rate of Total Crime against SCs (2023)'	Use the official definition and denominator.
2023 chargesheeting rate	CSV field 'Chargesheeting Rate (2023)'	Not a conviction rate; consult source notes.
SC population	CSV field referencing 2011, in lakhs	Reference year differs from the case-count years.

The derived change measures are created for descriptive comparison only. They do not adjust for population changes, reporting practices or other contextual differences.

Appendix D. Data Quality and Validation Log

Validation item	Observed/expected handling
Input dimensions	39 raw rows and 8 columns.
Jurisdiction rows	36 rows retained after excluding aggregate totals.
Year fields	2021, 2022 and 2023 are numeric in the supplied CSV.
Rate missingness	4 missing values in the 2023 crime-rate field.
Chargesheeting missingness	9 missing values in the 2023 chargesheeting-rate field.
Zero baseline	Percentage change is undefined where 2021 is zero.
Aggregate rows	Excluded from State/UT rankings and chart totals.
Notebook compatibility	Notebook expects a different schema and needs updating.

The report's calculations are reproducible from the supplied CSV with the stated filtering rules. Before formal publication, compare the resulting aggregate sums with the official source table and document any differences.

Appendix E. Suggested Power BI Measures

Measure name	Suggested definition	Display guidance
Cases 2021	Sum of 2021 for State/UT rows	Show as reported case count.
Cases 2022	Sum of 2022 for State/UT rows	Show as reported case count.
Cases 2023	Sum of 2023 for State/UT rows	Show as reported case count.
Change 2021–2023	Cases 2023 minus Cases 2021	Use diverging bar chart.
Percentage change	Change divided by Cases 2021	Blank for zero baseline.
Crime rate 2023	Source field value	Do not replace missing with zero.

Chargesheeting rate 2023	Source field value	Label separately from conviction rate.
State/UT selector	Filter by jurisdiction	Keep aggregate rows out of rankings.

Any DAX measures should be checked against the source CSV using the same aggregate-row rules. If a measure intentionally includes national totals, the report page should state that clearly and avoid summing a total row with its component rows.

Appendix F. Chart Interpretation Guide

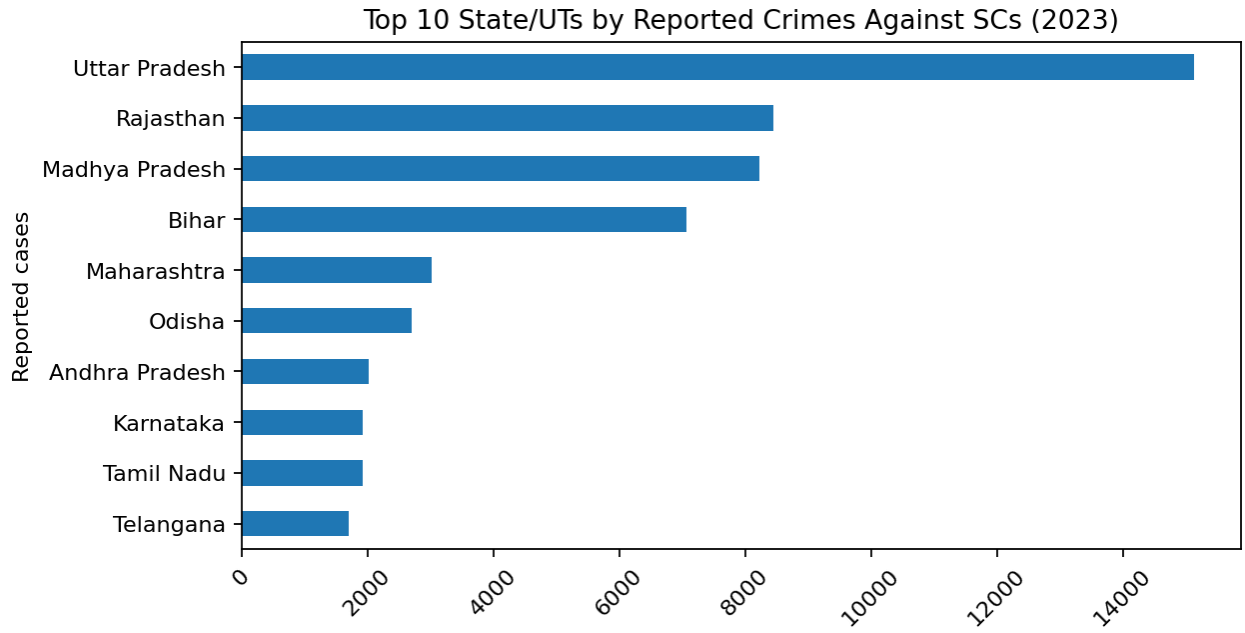


Figure A. Highest reported 2023 counts among State/UT rows.

The chart compares absolute reported counts. It does not normalize by population and should not be described as a risk ranking. The underlying State/UT table should be available to the reader so values can be checked.

Appendix G. Recommended Notebook Corrections

1. Change the input path to the uploaded NCRB CII Table 7A.1 CSV.

2. Replace references to the 2020 column with the actual available years 2021, 2022 and 2023.

3. Update the derived change fields to compare 2021 with 2023.

4. Rename population and rate fields using the actual SC-specific source labels.

5. Exclude aggregate rows using normalized labels while preserving them separately for validation.

6. Recheck the IQR calculations on the intended numeric field and state how outliers are used.

7. Run all cells from a clean kernel and verify the saved output columns.

8. Validate notebook totals and chart values against an independent CSV calculation.

Appendix H. Submission Review Checklist

• Cover title matches the submitted project.

• Student name and register number are correct.

• All charts show clear titles, units and year labels.

• Aggregate totals are not mixed into State/UT comparisons.

• Missing rate values remain distinguishable from zero.

• Notebook schema mismatch is disclosed and corrected before demonstration.

• Power BI report opens in the intended Power BI Desktop version.

• All reported numbers can be traced to the uploaded CSV.

• Source table definitions and footnotes are checked before final submission.

• Conclusions remain descriptive and do not claim causation.